

Pre-work · Integumentary System

Integumentary System. Read the Part A chapter first, then work this in order. This is the first organ where all four tissue types show up together, so it is where the module gets used rather than listed.

Module 1 · 4 of 4

Bring this to class

How to work this pre-work

The order matters more than the hours. Context first, content second. Most students who struggle did every piece of this, in the wrong sequence.

- 1 Read**
Read the Part A chapter first. This packet does not repeat it. It puts it to work.
- 2 Organize**
Panel 1. Turn the reading into mini visuals. Not copied sentences, your structure.
- 3 Draw**
Panel 2. Brain dump from memory, then draw, then correct in a second colour.
- 4 Apply**
Panel 3. Notes closed for the first pass. Open them only to check.
- 5 Retrieve**
Later in the week, not the same night. Cover everything and redraw your mini visuals from memory. What you cannot redraw is what you do not yet know.

Why this order. Reading before you organize gives you context. Organizing before you draw forces you to decide what connects to what. Drawing before you apply is where you find out what you actually know. If you skip to the application questions, you will answer them with the notes open and learn almost nothing.

PANEL 1 OF 3

Organize it

Turn the integumentary chapter into mini visuals. Seven chunks, and the ladder is the one you will use on every exam.

FIVE SHAPES, THAT IS THE WHOLE TOOLKIT

Information has a shape. Pick the wrong one and the visual is noise. Pick the right one and you can redraw it from memory in twenty seconds. Ladder for an ordered series. Chain for a sequence, with arrows. Split for a pair you keep confusing. Hub for one thing with parts hanging off it. Grid for two variables crossed.

Your chunks for this packet

Each chunk gets one visual, in the shape that fits it, drawn on the mini visual sheet at the end of this packet. Each is made once, so nothing here is done twice. If a chunk

PANEL 2 OF 3

Draw it

Panel 1 was the shape of the information. This is the structure itself, drawn from memory, then corrected.

HOW A BRAIN DUMP WORKS

Set a timer for two minutes. Close everything, write and sketch every piece you can recall on blank paper. Do not organize it. When the timer stops, then draw. The gap between the dump and the drawing is the part that teaches you.

PANEL 3 OF 3

Apply it

First pass with everything closed. Then open the notes in a second colour and mark what you missed.

WORKED EXAMPLE

A burn destroys the epidermis and the upper dermis but leaves hair follicles and glands intact. Predict whether it regenerates, and say why.

Answer. Yes, it regenerates, from the follicles and glands.

Reasoning. Ask one question of every burn: is there a source of new epidermal cells left inside the wound. The stratum basale is gone at the surface, but hair follicles and glands are lined with epidermal cells and they sit deep in the dermis, so they survive. New epithelium spreads outward from each surviving structure until the sheets meet. A full-thickness burn destroys those too, which is why it cannot heal from the middle and needs grafting.

takes longer than three minutes you are copying instead of organizing.

1 LADDER

The five strata

Five rungs, basale at the bottom. Beside each rung write what the keratinocyte is doing at that level. The mnemonic goes in the margin, not on the ladder.

2 HUB

The four cells of the epidermis

One hub, four spokes: keratinocyte, melanocyte, intraepidermal macrophage, Merkel cell. Percentage on one line, job on the next. Mark which stratum each lives in.

3 CHAIN

Melanin transfer

Melanocyte, arrow, projection between keratinocytes, arrow, granules enter the keratinocyte, arrow, granules cluster over the nucleus. Draw the last step, do not just write it. That veil is the whole point.

4 SPLIT

Papillary against reticular dermis

One split. Fraction of the dermis, fibres, and what lives in each on the branches. Add the three structures inside a dermal papilla.

5 GRID

The four skin glands

Sebaceous, eccrine, apocrine, ceruminous down the side. Where they are, what they secrete, where the duct opens, and when they switch on across the top.

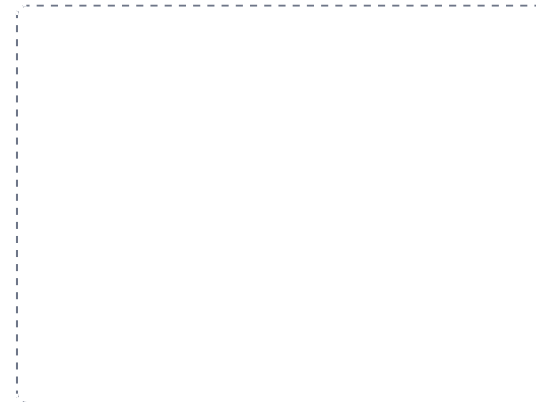
DRAWING 1

The skin in cross section

BRAIN DUMP FIRST

Two minutes. Every layer and every structure in the skin.

Draw the full thickness: epidermis, dermis with papillary and reticular regions, and hypodermis. Add a hair follicle running down, a sebaceous gland on it, an arrector pili muscle, an eccrine gland with its duct reaching the surface, dermal papillae with capillary loops and Meissner corpuscles, and a lamellated corpuscle in the hypodermis. Label everything.



In your notebook, head the page **M1-1 The skin in cross section** so you can find it later. Correct it in a second colour.

YOU KNOW IT WHEN

You can draw a horizontal line across your own drawing at any depth and say what a burn to that depth would destroy.

Set A · Layers and cells

1. Name the strata in order, deep to superficial, for thick skin.

2. Which stratum is missing in thin skin, and where on the body would you find thick skin?

3. The cell that alerts the immune system to an invader in the epidermis is the _____.

4. Which epidermal cell is least numerous, where does it sit, and what does it contact?

Set B · Colour and pigment

1. Two people have very different skin colour but the same number of melanocytes. Explain the difference.

2. A patient appears yellow in the skin and the sclera. Name the finding, the pigment, and the organ you would suspect.

3. A patient with dark skin is suspected of being cyanotic. Say where you would look instead, and why.

4. Explain why a tan is protective and why melanocytes are still damaged by the same exposure.

Set C · Structures and consequences

1. A gland opens onto a hair follicle in the axilla and switches on at puberty. Name it.

6 HUB

The nail

Nail at the centre. Spoke to body, free edge, root, lunula, hyponychium, eponychium, matrix. One line each on what it is or does.

7 CHAIN

Thermoregulation, both directions

Two chains from one starting point. Too warm goes one way, too cold the other. Put vessels and sweat on both chains and shivering on one.

YOU KNOW PANEL 1 WORKED WHEN

Someone hands you a blank page and you can redraw any one of these from memory in under a minute, explaining it out loud while you draw.

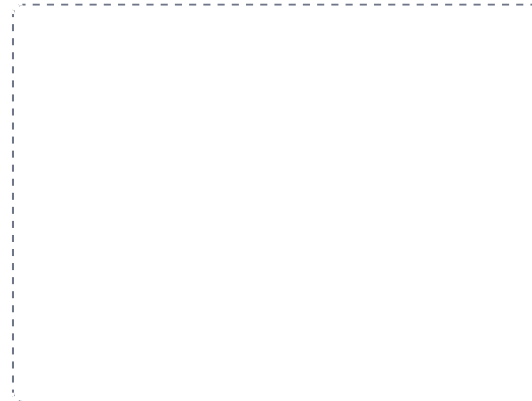
DRAWING 2

The five strata, cell by cell

BRAIN DUMP FIRST

One minute. The strata in order and what changes between them.

Zoom in on the epidermis alone. Draw the five strata stacked, with the cells changing shape as they rise: cuboidal and dividing at the base, spiny in the middle, granular above, clear, then flat dead squames at the top. Mark where the nucleus disappears and where lamellar granules release their sealant. Mark which layer is missing in thin skin.



In your notebook, head the page **M1-2 The five strata, cell by cell** so you can find it later. Correct it in a second colour.

YOU KNOW IT WHEN

You can explain why the epidermis survives without a blood supply, using only your drawing.

2. Damage to the nail matrix. Predict the effect on the nail and say why.

3. A patient loses the ability to sweat over a large area. Predict the consequence in hot weather and name the function lost.

4. Explain why a steroid cream works on inflamed skin but a large protein would not cross at all.

YOU KNOW PANEL 3 WORKED WHEN

Your first-pass answers and your corrections are in two different colours, and you can say out loud why each correction was a correction.

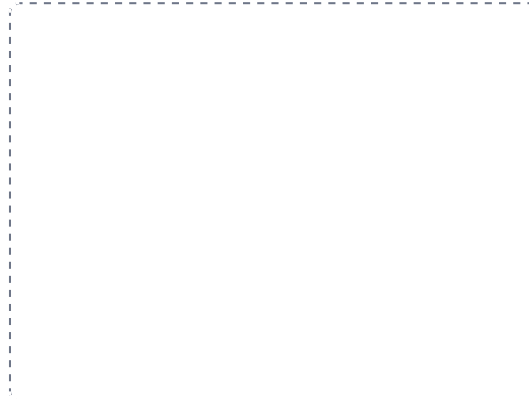
DRAWING 3

Hair follicle and bulb

BRAIN DUMP FIRST

One minute. The parts of a hair and everything attached to the follicle.

Draw one follicle from the surface down to the bulb. Show the shaft above the skin and the root below, the three concentric layers, the external and internal root sheaths, the dermal root sheath, and at the base the bulb with the papilla and the matrix. Attach the sebaceous gland, the arrector pili, and the hair root plexus.



In your notebook, head the page **M1-3 Hair follicle and bulb** so you can find it later. Correct it in a second colour.

YOU KNOW IT WHEN

You can say where the hair actually grows from and why cutting it does not change that.

Mini visual sheet

One frame per chunk, shaped for what goes in it. Tall frames are ladders, wide strips are chains, squares are hubs. Draw straight onto this sheet. If a frame is too small for your handwriting, redraw that one in your notes and put the page number on the line.

1 The five strata

LADDER

2 The four cells of the epidermis

HUB

3 Melanin transfer

CHAIN

4 Papillary against reticular dermis

SPLIT

5 The four skin glands

GRID

6 The nail

HUB



7 Thermoregulation, both directions

CHAIN

